

PG MITM Attack Simulation

Key Generation

Three GPG keys were created for:

- Alice
- Bob
- Mallory

gpg --full-generate-key

```
home/kali/.gnupg/pubring.kbx
ub  rsa2048 2025-12-03 [SC] [expired: 2025-12-04]
    5D5037B33A86F29DE26A9ACB44A7E336A1AB348F
id  [ expired] nino (123) <nino.repgmail>

ub  rsa4096 2026-05-25 [SC] [expires: 2027-05-25]
    CCC529DD92DFCDC33477A20895E2EE9BD4BFC7CE
id  [ultimate] nino (test123) <nino.testgmail.com>
ub  rsa4096 2026-05-25 [E] [expires: 2027-05-25]

ub  rsa4096 2026-05-25 [SC] [expires: 2027-05-25]
    663B4EA3B6A82074F140610AA45E3F42F60C933A
id  [ultimate] Alice (123) <alice@example.com>
ub  rsa4096 2026-05-25 [E] [expires: 2027-05-25]

ub  rsa4096 2026-05-25 [SC] [expires: 2027-05-25]
    DD26D0761937C9E80F29DAFB06244AEBA5E63CF4
id  [ultimate] bob (test) <bob@example.com>
ub  rsa4096 2026-05-25 [E] [expires: 2027-05-25]

ub  rsa4096 2026-05-25 [SC] [expires: 2027-05-25]
    14FB66B8408ADDB09B510DB2364B483B446E1694
id  [ultimate] mallorz (ja) <mallory@example.com>
ub  rsa4096 2026-05-25 [E] [expires: 2027-05-25]

—(kali@kali)~—
```

Fingerprint Verification

Fingerprints for all keys were displayed using:

gpg --fingerprint

Fingerprints are important because they uniquely identify a public key.

MITM Attack Simulation

Mallory exported her public key and renamed it as Bob's public key.

Alice imported the fake key and encrypted a message intended for Bob.

However, the message was encrypted using Mallory's key, allowing Mallory to decrypt the message successfully.

```
(kali㉿kali)-[~]
└─$ gpg --armor --export mallory@example.com > bob_pubkey.asc

(kali㉿kali)-[~]
└─$ gpg --import bob_pubkey.asc
gpg: key 364B483B446E1694: "mallorz (ja) <mallory@example.com>" not changed
gpg: Total number processed: 1
gpg:      unchanged: 1

(kali㉿kali)-[~]
└─$ echo "Confidential message for Bob" > secret.txt

(kali㉿kali)-[~]
└─$ gpg --encrypt --recipient bob@example.com secret.txt

(kali㉿kali)-[~]
└─$ gpg --decrypt secret.txt.gpg
gpg: encrypted with rsa4096 key, ID A8D2587E6F3D5406, created 2026-05-25
      "bob (test) <bob@example.com>"
gpg: WARNING: server 'gpg-agent' is older than us (2.4.8 < 2.4.9)
gpg: Note: Outdated servers may lack important security fixes.
gpg: Note: Use the command "gpgconf --kill all" to restart them.
Confidential message for Bob
```

Reflection

Why doesn't GPG detect MITM attacks automatically?

GPG cannot automatically verify whether a public key truly belongs to the claimed person.

What is a fingerprint and why is it important?

A fingerprint is a unique identifier of a public key. It helps verify the authenticity of the key.

Why is email not secure for exchanging keys?

An attacker can intercept or replace public keys during transmission.

How does the Web of Trust reduce MITM risks?

The Web of Trust allows users to verify and trust keys through trusted signatures and fingerprint verification.